DBMS Mini Project

Title: Legal Evidence Chain-of-Custody and Intelligent Case Investigation System

TEAM MEMBERS:

STUDENT 1;

NAME: Sindhu GV

SRN: PES2UG24CS661

STUDENT 2;

NAME: Vanishree

SRN: PES2UG24CS665

DML

```
mysql> show tables;
+---------------------------------------+
| Tables_in_legal_evidence_custody_db |
+---------------------------------------+
| belongs_to                            |
| case                                  |
| custody_transfer                      |
| evidence                              |
| evidence_category                     |
| evidence_summary                      |
| forensic_test                         |
| involves                              |
| location                              |
| person                                |
| user                                  |
+---------------------------------------+
11 rows in set (0.00 sec)

mysql>
```

Insert

```
mysql> INSERT INTO evidence_category
    -> (Category_Name, Category_Description)
    -> VALUES
    -> ('Digital', 'Digital evidence such as mobile phones and CCTV footage'),
    -> ('Physical', 'Physical objects collected from a crime scene'),
    -> ('Biological', 'Biological samples collected for forensic analysis'),
    -> ('Documentary', 'Documents and written records');
Query OK, 4 rows affected (0.27 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM evidence_category;
+-------------+---------------+-----------------------------------------------------------+
| Category_ID | Category_Name | Category_Description                                      |
+-------------+---------------+-----------------------------------------------------------+
|           1 | Digital       | Digital evidence such as mobile phones and CCTV footage |
|           2 | Physical      | Physical objects collected from a crime scene           |
|           3 | Biological    | Biological samples collected for forensic analysis      |
|           4 | Documentary   | Documents and written records                           |
+-------------+---------------+-----------------------------------------------------------+
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO location
    -> (Location_Name, Location_Type, Address, City, State, Country)
    -> VALUES
    -> ('Central Police Station', 'Police Station', 'MG Road', 'Bengaluru', 'Karnataka', 'India'),
    -> ('Evidence Storage Room', 'Storage', 'Central Police Station', 'Bengaluru', 'Karnataka', 'India'),
    -> ('Forensic Science Laboratory', 'Forensic Lab', 'Electronic City', 'Bengaluru', 'Karnataka', 'India'),
    -> ('District Court', 'Court', 'Court Road', 'Bengaluru', 'Karnataka', 'India');
Query OK, 4 rows affected (0.14 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO `user`
    -> (Name, Role, Department, Phone_Number, Email, Password_Hash, Status, Location_ID)
    -> VALUES
    -> ('Ravi Kumar', 'Officer', 'Investigation', '9876543210',
    ->  'ravi@chainsecure.com', 'HASH_RAVI_123', 'Active', 1),
    ->
    -> ('Priya Sharma', 'Forensic Officer', 'Forensics', '9876543211',
    ->  'priya@chainsecure.com', 'HASH_PRIYA_123', 'Active', 3),
    ->
    -> ('Arjun Rao', 'Officer', 'Investigation', '9876543212',
    ->  'arjun@chainsecure.com', 'HASH_ARJUN_123', 'Active', 1),
    ->
    -> ('Meera Singh', 'Admin', 'Administration', '9876543213',
    ->  'meera@chainsecure.com', 'HASH_MEERA_123', 'Active', 1);
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT User_ID, Name, Role, Department, Status
    -> FROM `user`;
+---------+-------------+------------------+----------------+--------+
| User_ID | Name        | Role             | Department     | Status |
+---------+-------------+------------------+----------------+--------+
|       1 | Ravi Kumar  | Officer          | Investigation  | Active |
|       2 | Priya Sharma | Forensic Officer | Forensics      | Active |
|       3 | Arjun Rao   | Officer          | Investigation  | Active |
|       4 | Meera Singh | Admin            | Administration | Active |
+---------+-------------+------------------+----------------+--------+
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO `case`
    -> (Case_Number, Title, Description, Case_Type, Opened_Date, Status)
    -> VALUES
    -> ('CASE-001',
    ->  'Jewellery Store Robbery',
    ->  'Investigation of a robbery at a jewellery store',
    ->  'Robbery',
    ->  '2026-09-01',
    ->  'Open'),
    ->
    -> ('CASE-002',
    ->  'Vehicle Theft',
    ->  'Investigation of a stolen vehicle',
    ->  'Theft',
    ->  '2026-09-05',
    ->  'Open'),
    ->
    -> ('CASE-003',
    ->  'Online Financial Fraud',
    ->  'Investigation of an online financial fraud complaint',
    ->  'Cyber Crime',
    ->  '2026-09-10',
    ->  'Under Investigation');
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO person
    -> (Name, Person_Type, Contact_Info, Date_of_Birth, Identification_No)
    -> VALUES
    -> ('Rahul Verma', 'Witness', '9876500001', '1995-05-12', 'ID1001'),
    ->
    -> ('Suresh Reddy', 'Suspect', '9876500002', '1990-08-20', 'ID1002'),
    ->
    -> ('Anita Rao', 'Victim', '9876500003', '1988-02-15', 'ID1003'),
    ->
    -> ('Kiran Das', 'Witness', '9876500004', '1998-11-25', 'ID1004'),
    ->
    -> ('Vijay Kumar', 'Suspect', '9876500005', '1992-03-10', 'ID1005');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO evidence
    -> (Case_ID, Category_ID, Description, Collected_Date,
    ->  Collected_By, Current_Status, Storage_Location,
    ->  Hash_Value, Hash_Algorithm)
    -> VALUES
    -> (1, 1,
    ->  'Mobile phone recovered from crime scene',
    ->  '2026-09-01', 1, 'Stored',
    ->  'Evidence Storage Room',
    ->  'HASH001',
    ->  'SHA256'),
    ->
    -> (1, 2,
    ->  'Metal tool recovered near entrance',
    ->  '2026-09-01', 1, 'In_Lab',
    ->  'Forensic Science Laboratory',
    ->  'HASH002',
    ->  'SHA256'),
    ->
    -> (1, 1,
    ->  'CCTV footage from jewellery store',
    ->  '2026-09-02', 3, 'Stored',
    ->  'Evidence Storage Room',
    ->  'HASH003',
    ->  'SHA256'),
    ->
    -> (2, 2,
    ->  'Vehicle key recovered from suspect',
    ->  '2026-09-05', 3, 'Collected',
    ->  'Central Police Station',
    ->  'HASH004',
    ->  'SHA256'),
    ->
    -> (2, 1,
    ->  'GPS device recovered from vehicle',
    ->  '2026-09-06', 3, 'Stored',
    ->  'Evidence Storage Room',
    ->  'HASH005',
    ->  'SHA256'),
    ->
    -> (3, 4,
    ->  'Bank transaction document',
    ->  '2026-09-10', 1, 'Stored',
    ->  'Evidence Storage Room',
    ->  'HASH006',
    ->  'SHA256');
Query OK, 6 rows affected (0.03 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO forensic_test
    -> (Evidence_ID, Test_Type, Conducted_By, Test_Date, Result, Report_Path)
    -> VALUES
    -> (1, 'Mobile Data Extraction', 2,
    ->  '2026-09-03',
    ->  'Relevant communication records recovered',
    ->  '/reports/test001.pdf'),
    ->
    -> (2, 'Fingerprint Analysis', 2,
    ->  '2026-09-03',
    ->  'Partial fingerprint detected',
    ->  '/reports/test002.pdf'),
    ->
    -> (3, 'Video Analysis', 2,
    ->  '2026-09-04',
    ->  'Suspected individual identified in footage',
    ->  '/reports/test003.pdf'),
    ->
    -> (5, 'GPS Data Analysis', 2,
    ->  '2026-09-07',
    ->  'Vehicle movement history recovered',
    ->  '/reports/test004.pdf');
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO custody_transfer
    -> (Evidence_ID, Transfer_No, From_User_ID, To_User_ID,
    ->  Transfer_DateTime, Received_Time, From_Location_ID, Remarks)
    -> VALUES
    -> (1, 1, 1, 4,
    ->  '2026-09-01 10:00:00',
    ->  '2026-09-01 10:30:00',
    ->  1,
    ->  'Evidence submitted to storage'),
    ->
    -> (1, 2, 4, 2,
    ->  '2026-09-03 09:00:00',
    ->  '2026-09-03 09:30:00',
    ->  2,
    ->  'Sent for forensic examination'),
    ->
    -> (1, 3, 2, 4,
    ->  '2026-09-04 16:00:00',
    ->  '2026-09-04 16:30:00',
    ->  3,
    ->  'Returned after forensic examination'),
    ->
    -> (2, 1, 1, 2,
    ->  '2026-09-02 09:00:00',
    ->  '2026-09-02 09:30:00',
    ->  1,
    ->  'Fingerprint examination'),
    ->
    -> (3, 1, 3, 4,
    ->  '2026-09-02 11:00:00',
    ->  '2026-09-02 11:30:00',
    ->  1,
    ->  'CCTV secured');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

SELECT

```
mysql> SELECT * FROM evidence;
+-------------+---------+-------------+---------------------------------------+----------------+--------------+----------------+-----------------------------+------------+----------------+---------------+
| Evidence_ID | Case_ID | Category_ID | Description                           | Collected_Date | Collected_By | Current_Status | Storage_Location            | Hash_Value | Hash_Algorithm | Chain_Verified |
+-------------+---------+-------------+---------------------------------------+----------------+--------------+----------------+-----------------------------+------------+----------------+---------------+
|           1 |       1 |           1 | Mobile phone recovered from crime scene | 2026-09-01     |            1 | Stored         | Evidence Storage Room       | HASH001    | SHA256         |             0 |
|           2 |       1 |           2 | Metal tool recovered near entrance    | 2026-09-01     |            1 | In_Lab         | Forensic Science Laboratory | HASH002    | SHA256         |             0 |
|           3 |       1 |           1 | CCTV footage from jewellery store     | 2026-09-02     |            3 | Stored         | Evidence Storage Room       | HASH003    | SHA256         |             0 |
|           4 |       2 |           2 | Vehicle key recovered from suspect    | 2026-09-05     |            3 | Collected      | Central Police Station      | HASH004    | SHA256         |             0 |
|           5 |       2 |           1 | GPS device recovered from vehicle     | 2026-09-06     |            3 | Stored         | Evidence Storage Room       | HASH005    | SHA256         |             0 |
|           6 |       3 |           4 | Bank transaction document             | 2026-09-10     |            1 | Stored         | Evidence Storage Room       | HASH006    | SHA256         |             0 |
+-------------+---------+-------------+---------------------------------------+----------------+--------------+----------------+-----------------------------+------------+----------------+---------------+
6 rows in set (0.00 sec)

mysql> SELECT Evidence_ID, Description, Current_Status
    -> FROM evidence;
+-------------+---------------------------------------+----------------+
| Evidence_ID | Description                           | Current_Status |
+-------------+---------------------------------------+----------------+
|           1 | Mobile phone recovered from crime scene | Stored         |
|           2 | Metal tool recovered near entrance    | In_Lab         |
|           3 | CCTV footage from jewellery store     | Stored         |
|           4 | Vehicle key recovered from suspect    | Collected      |
|           5 | GPS device recovered from vehicle     | Stored         |
|           6 | Bank transaction document             | Stored         |
+-------------+---------------------------------------+----------------+
6 rows in set (0.00 sec)
```

Where Clause

```
mysql> SELECT *
    -> FROM evidence
    -> WHERE Current_Status = 'Stored';
+-------------+---------+-------------+---------------------------------------+----------------+--------------+----------------+-----------------------------+------------+----------------+---------------+
| Evidence_ID | Case_ID | Category_ID | Description                           | Collected_Date | Collected_By | Current_Status | Storage_Location            | Hash_Value | Hash_Algorithm | Chain_Verified |
+-------------+---------+-------------+---------------------------------------+----------------+--------------+----------------+-----------------------------+------------+----------------+---------------+
|           1 |       1 |           1 | Mobile phone recovered from crime scene | 2026-09-01     |            1 | Stored         | Evidence Storage Room       | HASH001    | SHA256         |             0 |
|           3 |       1 |           1 | CCTV footage from jewellery store     | 2026-09-02     |            3 | Stored         | Evidence Storage Room       | HASH003    | SHA256         |             0 |
|           5 |       2 |           1 | GPS device recovered from vehicle     | 2026-09-06     |            3 | Stored         | Evidence Storage Room       | HASH005    | SHA256         |             0 |
|           6 |       3 |           4 | Bank transaction document             | 2026-09-10     |            1 | Stored         | Evidence Storage Room       | HASH006    | SHA256         |             0 |
+-------------+---------+-------------+---------------------------------------+----------------+--------------+----------------+-----------------------------+------------+----------------+---------------+
4 rows in set (0.00 sec)
```

Order by

```
mysql> SELECT Evidence_ID, Description, Collected_Date
    -> FROM evidence
    -> ORDER BY Collected_Date DESC;
+-------------+---------------------------------------+----------------+
| Evidence_ID | Description                           | Collected_Date |
+-------------+---------------------------------------+----------------+
|           6 | Bank transaction document             | 2026-09-10     |
|           5 | GPS device recovered from vehicle     | 2026-09-06     |
|           4 | Vehicle key recovered from suspect    | 2026-09-05     |
|           3 | CCTV footage from jewellery store     | 2026-09-02     |
|           1 | Mobile phone recovered from crime scene | 2026-09-01     |
|           2 | Metal tool recovered near entrance    | 2026-09-01     |
+-------------+---------------------------------------+----------------+
6 rows in set (0.00 sec)
```

Join

```
mysql> SELECT
    ->     e.Evidence_ID,
    ->     c.Case_Number,
    ->     c.Title,
    ->     ec.Category_Name,
    ->     e.Description,
    ->     e.Current_Status
    -> FROM evidence e
    -> JOIN `case` c
    ->     ON e.Case_ID = c.Case_ID
    -> JOIN evidence_category ec
    ->     ON e.Category_ID = ec.Category_ID;
+-------------+-------------+--------------------------+---------------+---------------------------------------+----------------+
| Evidence_ID | Case_Number | Title                    | Category_Name | Description                           | Current_Status |
+-------------+-------------+--------------------------+---------------+---------------------------------------+----------------+
|           1 | CASE-001    | Jewellery Store Robbery  | Digital       | Mobile phone recovered from crime scene | Stored         |
|           2 | CASE-001    | Jewellery Store Robbery  | Physical      | Metal tool recovered near entrance    | In_Lab         |
|           3 | CASE-001    | Jewellery Store Robbery  | Digital       | CCTV footage from jewellery store     | Stored         |
|           4 | CASE-002    | Vehicle Theft            | Physical      | Vehicle key recovered from suspect    | Collected      |
|           5 | CASE-002    | Vehicle Theft            | Digital       | GPS device recovered from vehicle     | Stored         |
|           6 | CASE-003    | Online Financial Fraud   | Documentary   | Bank transaction document             | Stored         |
+-------------+-------------+--------------------------+---------------+---------------------------------------+----------------+
6 rows in set (0.00 sec)

mysql> SELECT
    ->     e.Evidence_ID,
    ->     e.Description,
    ->     u.Name AS Collected_By,
    ->     u.Role
    -> FROM evidence e
    -> JOIN `user` u
    ->     ON e.Collected_By = u.User_ID;
+-------------+---------------------------------------+--------------+---------+
| Evidence_ID | Description                           | Collected_By | Role    |
+-------------+---------------------------------------+--------------+---------+
|           1 | Mobile phone recovered from crime scene | Ravi Kumar   | Officer |
|           2 | Metal tool recovered near entrance    | Ravi Kumar   | Officer |
|           3 | CCTV footage from jewellery store     | Arjun Rao    | Officer |
|           4 | Vehicle key recovered from suspect    | Arjun Rao    | Officer |
|           5 | GPS device recovered from vehicle     | Arjun Rao    | Officer |
|           6 | Bank transaction document             | Ravi Kumar   | Officer |
+-------------+---------------------------------------+--------------+---------+
6 rows in set (0.00 sec)
```

Group by

```
mysql> SELECT
    ->     ec.Category_Name,
    ->     COUNT(e.Evidence_ID) AS Total_Evidence
    -> FROM evidence e
    -> JOIN evidence_category ec
    ->     ON e.Category_ID = ec.Category_ID
    -> GROUP BY ec.Category_Name;
+---------------+----------------+
| Category_Name | Total_Evidence |
+---------------+----------------+
| Digital       |              3 |
| Documentary   |              1 |
| Physical      |              2 |
+---------------+----------------+
3 rows in set (0.00 sec)
```

Having

```
mysql> SELECT
    ->     ec.Category_Name,
    ->     COUNT(e.Evidence_ID) AS Total_Evidence
    -> FROM evidence e
    -> JOIN evidence_category ec
    ->     ON e.Category_ID = ec.Category_ID
    -> GROUP BY ec.Category_Name
    -> HAVING COUNT(e.Evidence_ID) > 1;
+---------------+----------------+
| Category_Name | Total_Evidence |
+---------------+----------------+
| Digital       |              3 |
| Physical      |              2 |
+---------------+----------------+
2 rows in set (0.00 sec)
```

Count

```
mysql> SELECT COUNT(*) AS Total_Evidence
    -> FROM evidence;
+----------------+
| Total_Evidence |
+----------------+
|              6 |
+----------------+
1 row in set (0.00 sec)
```

Min

```
mysql> SELECT MIN(Collected_Date) AS First_Evidence_Date
    -> FROM evidence;
+---------------------+
| First_Evidence_Date |
+---------------------+
| 2026-09-01          |
+---------------------+
1 row in set (0.00 sec)
```

Max

```
mysql> SELECT MAX(Collected_Date) AS Latest_Evidence_Date
    -> FROM evidence;
+----------------------+
| Latest_Evidence_Date |
+----------------------+
| 2026-09-10           |
+----------------------+
1 row in set (0.00 sec)
```

MIN,MAX,Count

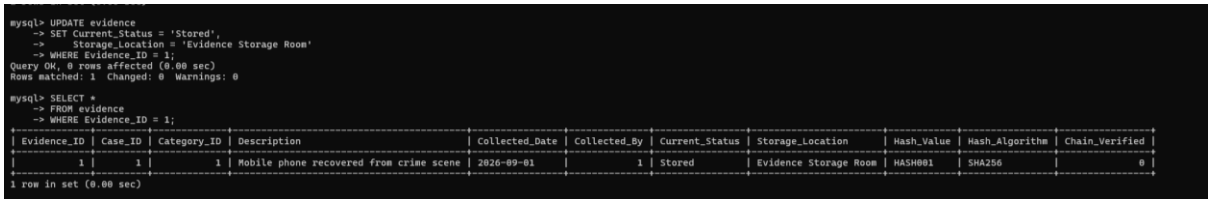
```
mysql> SELECT
    ->     COUNT(*) AS Total_Evidence,
    ->     MIN(Collected_Date) AS First_Collection,
    ->     MAX(Collected_Date) AS Latest_Collection
    -> FROM evidence;
+----------------+------------------+-------------------+
| Total_Evidence | First_Collection | Latest_Collection |
+----------------+------------------+-------------------+
|              6 | 2026-09-01       | 2026-09-10        |
+----------------+------------------+-------------------+
1 row in set (0.00 sec)
```

Nested Query

```
mysql> SELECT Evidence_ID, Description
    -> FROM evidence
    -> WHERE Case_ID IN (
    ->     SELECT Case_ID
    ->     FROM `case`
    ->     WHERE Status = 'Open'
    -> );
+-------------+-----------------------------------------+
| Evidence_ID | Description                             |
+-------------+-----------------------------------------+
|           1 | Mobile phone recovered from crime scene |
|           2 | Metal tool recovered near entrance      |
|           3 | CCTV footage from jewellery store       |
|           4 | Vehicle key recovered from suspect      |
|           5 | GPS device recovered from vehicle       |
+-------------+-----------------------------------------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT Name
    -> FROM `user`
    -> WHERE User_ID IN (
    ->     SELECT DISTINCT Collected_By
    ->     FROM evidence
    -> );
+-----------+
| Name      |
+-----------+
| Ravi Kumar |
| Arjun Rao  |
+-----------+
2 rows in set (0.00 sec)
```

Update



```
mysql> UPDATE evidence
    -> SET Current_Status = 'Stored',
    ->     Storage_Location = 'Evidence Storage Room'
    -> WHERE Evidence_ID = 1;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql> SELECT *
    -> FROM evidence
    -> WHERE Evidence_ID = 1;
+-------------+---------+-------------+-----------------------------------+----------------+-------------+----------------+-----------+----------------+----------------+
| Evidence_ID | Case_ID | Category_ID | Description                       | Collected_Date | Collected_By | Current_Status | Storage_Location      | Hash_Value | Hash_Algorithm | Chain_Verified |
+-------------+---------+-------------+-----------------------------------+----------------+-------------+----------------+-----------+----------------+----------------+
|           1 |       1 |           1 | Mobile phone recovered from crime scene | 2026-09-01     |           1 | Stored         | Evidence Storage Room | HASH001    | SHA256         |              0 |
+-------------+---------+-------------+-----------------------------------+----------------+-------------+----------------+-----------+----------------+----------------+
1 row in set (0.00 sec)
```

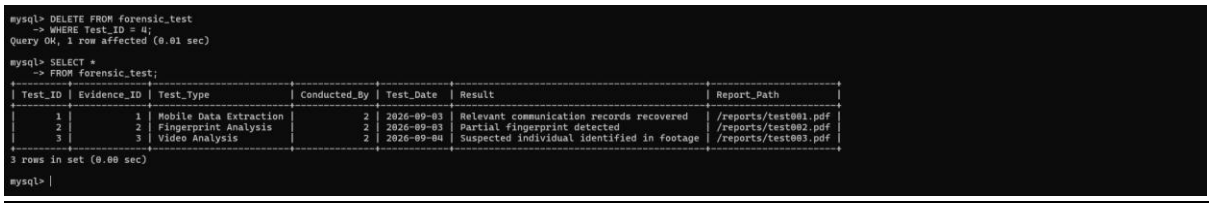
```
mysql> UPDATE evidence
    -> SET Current_Status = 'In_Lab'
    -> WHERE Evidence_ID = 1;
Query OK, 1 row affected (0.09 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT Evidence_ID, Description, Current_Status
    -> FROM evidence
    -> WHERE Evidence_ID = 1;
+-------------+---------------------------------------+----------------+
| Evidence_ID | Description                           | Current_Status |
+-------------+---------------------------------------+----------------+
|           1 | Mobile phone recovered from crime scene | In_Lab         |
+-------------+---------------------------------------+----------------+
1 row in set (0.01 sec)

mysql> UPDATE evidence
    -> SET Current_Status = 'Stored'
    -> WHERE Evidence_ID = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
```

Delete



```
mysql> DELETE FROM forensic_test
    -> WHERE Test_ID = 4;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT *
    -> FROM forensic_test;
+---------+-------------+-----------------------+--------------+------------+-----------------------------------------+-----------------------+
| Test_ID | Evidence_ID | Test_Type             | Conducted_By | Test_Date  | Result                                  | Report_Path           |
+---------+-------------+-----------------------+--------------+------------+-----------------------------------------+-----------------------+
|       1 |           1 | Mobile Data Extraction |            2 | 2026-09-03 | Relevant communication records recovered | /reports/test001.pdf |
|       2 |           2 | Fingerprint Analysis  |            2 | 2026-09-03 | Partial fingerprint detected            | /reports/test002.pdf |
|       3 |           3 | Video Analysis        |            2 | 2026-09-04 | Suspected individual identified in footage | /reports/test003.pdf |
+---------+-------------+-----------------------+--------------+------------+-----------------------------------------+-----------------------+
3 rows in set (0.00 sec)

mysql>
```

Advanced SQL

TRANSACTION — Evidence Custody Transfer

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> INSERT INTO custody_transfer
    -> (
    ->     Evidence_ID,
    ->     Transfer_No,
    ->     From_User_ID,
    ->     To_User_ID,
    ->     Transfer_DateTime,
    ->     Received_Time,
    ->     From_Location_ID,
    ->     Remarks
    -> )
    -> VALUES
    -> (
    ->     2,
    ->     2,
    ->     1,
    ->     2,
    ->     '2026-09-28 10:00:00',
    ->     '2026-09-28 10:30:00',
    ->     1,
    ->     'Evidence transferred to forensic officer'
    -> );
Query OK, 1 row affected (0.06 sec)

mysql>
mysql> UPDATE evidence
    -> SET Current_Status = 'In_Lab'
    -> WHERE Evidence_ID = 2;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT *
    -> FROM custody_transfer
    -> WHERE Evidence_ID = 2;
+-------------+-------------+---------------+-------------+---------------------+---------------------+-----------------+-----------------------------------------+
| Evidence_ID | Transfer_No | From_User_ID | To_User_ID | Transfer_DateTime   | Received_Time       | From_Location_ID | Remarks                                 |
+-------------+-------------+---------------+-------------+---------------------+---------------------+-----------------+-----------------------------------------+
|           2 |           1 |            1 |           2 | 2026-09-02 09:00:00 | 2026-09-02 09:30:00 |               1 | Fingerprint examination                 |
|           2 |           2 |            1 |           2 | 2026-09-28 10:00:00 | 2026-09-28 10:30:00 |               1 | Evidence transferred to forensic officer |
+-------------+-------------+---------------+-------------+---------------------+---------------------+-----------------+-----------------------------------------+
2 rows in set (0.01 sec)

mysql> SELECT Evidence_ID, Current_Status
    -> FROM evidence
    -> WHERE Evidence_ID = 2;
+-------------+----------------+
| Evidence_ID | Current_Status |
+-------------+----------------+
|           2 | In_Lab         |
+-------------+----------------+
1 row in set (0.00 sec)

mysql>
```

TRIGGER — Automatically Update Evidence Status

```
mysql> DELIMITER //
mysql> CREATE TRIGGER trg_after_custody_transfer
    -> AFTER INSERT ON custody_transfer
    -> FOR EACH ROW
    -> BEGIN
    ->     UPDATE evidence
    ->     SET Current_Status = 'In_Transit',
    ->         Chain_Verified = FALSE
    ->     WHERE Evidence_ID = NEW.Evidence_ID;
    -> END//
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> SHOW TRIGGERS;
+-----------------------------+--------+-----------------+----------------------------
```

```
mysql> INSERT INTO custody_transfer
    -> (
    ->     Evidence_ID,
    ->     Transfer_No,
    ->     From_User_ID,
    ->     To_User_ID,
    ->     Transfer_DateTime,
    ->     Received_Time,
    ->     From_Location_ID,
    ->     Remarks
    -> )
    -> VALUES
    -> (
    ->     3,
    ->     2,
    ->     1,
    ->     2,
    ->     '2026-09-28 11:00:00',
    ->     '2026-09-28 11:30:00',
    ->     1,
    ->     'CCTV footage sent for forensic analysis'
    -> );
Query OK, 1 row affected (0.03 sec)

mysql> SELECT Evidence_ID, Current_Status, Chain_Verified
    -> FROM evidence
    -> WHERE Evidence_ID = 3;
+-------------+----------------+----------------+
| Evidence_ID | Current_Status | Chain_Verified |
+-------------+----------------+----------------+
|           3 | In_Transit     |              0 |
+-------------+----------------+----------------+
1 row in set (0.00 sec)

mysql>
```

STORED FUNCTION — Verify Custody Chain

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION verify_chain_integrity(p_evidence_id INT)
    -> RETURNS VARCHAR(20)
    -> DETERMINISTIC
    -> READS SQL DATA
    -> BEGIN
    ->     DECLARE invalid_count INT DEFAULT 0;
    ->
    ->     SELECT COUNT(*)
    ->     INTO invalid_count
    ->     FROM custody_transfer ct
    ->     WHERE ct.Evidence_ID = p_evidence_id
    ->       AND (
    ->           ct.Received_Time < ct.Transfer_DateTime
    ->           OR
    ->           (
    ->               ct.Transfer_No > 1
    ->               AND NOT EXISTS (
    ->                   SELECT 1
    ->                   FROM custody_transfer previous_transfer
    ->                   WHERE previous_transfer.Evidence_ID = ct.Evidence_ID
    ->                     AND previous_transfer.Transfer_No = ct.Transfer_No - 1
    ->               )
    ->           )
    ->       );
    ->
    ->     IF invalid_count = 0 THEN
    ->         RETURN 'VALID';
    ->     ELSE
    ->         RETURN 'INVALID';
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.10 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> SELECT verify_chain_integrity(1) AS Chain_Status;
+--------------+
| Chain_Status |
+--------------+
| VALID        |
+--------------+
1 row in set (0.05 sec)

mysql> SELECT verify_chain_integrity(2) AS Chain_Status;
+--------------+
| Chain_Status |
+--------------+
| VALID        |
+--------------+
1 row in set (0.00 sec)

mysql>
```

VIEW — Evidence Investigation Summary

```
mysql> SHOW FULL TABLES
    -> WHERE Table_type = 'VIEW';
+---------------------------------------+------------+
| Tables_in_legal_evidence_custody_db | Table_type |
+---------------------------------------+------------+
| evidence_summary                      | VIEW       |
+---------------------------------------+------------+
1 row in set (0.01 sec)

mysql> SELECT *
    -> FROM evidence_summary;
+-------------+-------------+-------------------------+---------------+-----------------------------------------+----------------+-----------------+
| Evidence_ID | Case_Number | Case_Title              | Category_Name | Description                             | Current_Status | Chain_Verified |
+-------------+-------------+-------------------------+---------------+-----------------------------------------+----------------+-----------------+
|           1 | CASE-001    | Jewellery Store Robbery | Digital       | Mobile phone recovered from crime scene | Stored         |               0 |
|           2 | CASE-001    | Jewellery Store Robbery | Physical      | Metal tool recovered near entrance      | In_Lab         |               0 |
|           3 | CASE-001    | Jewellery Store Robbery | Digital       | CCTV footage from jewellery store       | In_Transit    |               0 |
|           4 | CASE-002    | Vehicle Theft           | Physical      | Vehicle key recovered from suspect      | Collected      |               0 |
|           5 | CASE-002    | Vehicle Theft           | Digital       | GPS device recovered from vehicle       | Stored         |               0 |
|           6 | CASE-003    | Online Financial Fraud  | Documentary   | Bank transaction document               | Stored         |               0 |
+-------------+-------------+-------------------------+---------------+-----------------------------------------+----------------+-----------------+
6 rows in set (0.02 sec)

mysql>
```

```
mysql> SHOW CREATE VIEW evidence_summary;
+------------------+-----------------------------------------
```